Lecture Slides: Photosynthesis

• SLIDE 1 - PHOTOSYNTHESIS**

• Subtitle:* How plants turn sunlight, water and carbon-dioxide into food and oxygen.

• Key ideas**

• Energy change inside living cells

• How the parts of a chloroplast work together

• Why photosynthesis matters to people and the planet

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• SLIDE 2 - LEARNING OBJECTIVES**

By the end of this lesson you will be able to:

1. **Define** photosynthesis and name its two main stages in plain language.

2. **Label** the parts of a chloroplast and say where each stage happens.

3. **Explain** how light energy makes the carriers **ATP** and **NADPH**.

4. **Compare** C■, C■ and CAM pathways and give one reason each is useful.

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• SLIDE 3 - WHY PHOTOSYNTHESIS MATTERS**

• All green plants, many algae and cyanobacteria perform photosynthesis.

• Overall equation (balanced): **6 CO■ + 6 H■O + light -> C■H■■O■ + 6 O■**

• Sunlight is captured by pigments (chlorophyll a, chlorophyll b, carotenoids) - think of them as tiny solar panels.

• The sugar (glucose) made by plants feeds most food webs; the oxygen released lets us breathe.

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• SLIDE 4 - THE CHLOROPLAST: A MINI-FACTORY**

• **Double membrane** surrounds the organelle - outer and inner wall.

• Inside are **stacks of flattened sacs** called **thylakoids**; a stack = **grana**.

• Space inside a thylakoid = **lumen** (where light reactions happen).

• Fluid surrounding the thylakoids = **stroma** (site of the Calvin cycle).

• Chloroplasts have their own DNA and ribosomes - they can make some of their own proteins.

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• SLIDE 5 - LIGHT-DEPENDENT REACTIONS (ENERGY-CAPTURE STAGE)**

1. **Photon absorption** - pigments grab light energy and lift electrons to a higher level.

2. **Electron transport chain (ETC)** - excited electrons move through a series of carriers (like a relay).

3. **Proton pumping** - as electrons travel, H■ ions are pushed into the thylakoid lumen, creating a gradient.

4. **Chemiosmosis** - protons flow back through **ATP synthase** (a tiny turbine) and make **ATP**.

5. **NADP■ reduction** - the last electrons combine with NADP■ and H■ to form **NADPH**.

6. **Photolysis of water** - water splits to replace lost electrons; O■ is released as a by-product.

• Key terms (on-slide callouts):*

• **Photon** - a packet of light energy

• **ATP** - cell's energy "currency"

• **NADPH** - carrier that brings electrons to the next stage

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• SLIDE 6 - LIGHT-INDEPENDENT REACTIONS (CALVIN-BENSEN CYCLE)**

• Takes place in the **stroma** using ATP and NADPH from the light stage.

• **Step 1 - Carbon fixation:** enzyme **Rubisco** attaches CO■ to a 5-carbon sugar (**RuBP**) -> unstable 6-carbon molecule -> splits into two 3-carbon compounds.

• **Step 2 - Reduction:** each 3-carbon piece receives electrons from NADPH and a phosphate from ATP, becoming **G3P** (glyceraldehyde-3-phosphate).

• **Step 3 - Regeneration:** some G3P is recycled to rebuild RuBP, allowing the cycle to continue.

• **Output:** for every 6 CO■ fixed, the cycle uses ~18 ATP and 12 NADPH and makes about 2-3 glucose molecules.

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• SLIDE 7 - C■, C■ AND CAM PLANTS (ADAPTING TO ENVIRONMENT)**

| Feature | C■ (most plants) | C■ (e.g., maize, sugarcane) | CAM (e.g., cactus) |

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| Where CO■ is fixed | Directly in leaf cells (Calvin cycle) | First in **mesophyll** cells -> CO■ turned into a 4-C acid -> moved to **bundle-sheath** cells for Calvin cycle | CO■ taken at night, stored as malic acid, used by Calvin cycle during the day |

| Advantage | Simple, works well in cool, moist climates | Concentrates CO■, reduces wasteful **photorespiration** in hot, sunny conditions | Saves water by opening stomata at night |

| Trade-off | High photorespiration when it's hot | Requires extra energy to move carbon between cells | Slower growth because photosynthesis is limited to daylight hours |

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• SLIDE 8 - REAL-WORLD APPLICATIONS**

• **Crop improvement:** Breeding C■ traits into C■ crops could boost yields in warm regions.

• **Algal bio-fuels:** Fast-growing algae turn sunlight into oil that can replace petroleum.

• **Carbon sequestration:** Forests and re-forestation projects store CO■ captured through photosynthesis.

• **Synthetic biology:** Scientists are trying to copy light-harvesting complexes to make solar-energy devices.

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• SLIDE 9 - CRITICAL-THINKING QUESTIONS**

1. If atmospheric CO■ drops, how would the photosynthesis rate of a C■ plant compare to a C■ plant? Why?

2. How might higher temperatures increase photorespiration in temperate crops?

3. Why is the CAM pathway better than the C■ pathway in a desert, and what does the plant give up to use CAM?

4. A mutation makes Rubisco bind CO■ less well. Predict the effects on plant growth and on the food chain.

5. Imagine a solar panel that works like a chloroplast's light-dependent reactions. What part of the plant process would you try to copy, and what benefit could it give renewable-energy technology?

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• SLIDE 10 - SUMMARY & DESIGN CHALLENGE**

• Take-away points**

• Photosynthesis changes light energy into chemical energy (ATP + NADPH) inside chloroplasts.

• Light-dependent reactions happen in thylakoid membranes; the Calvin cycle runs in the stroma.

• C■, C■ and CAM plants use different tricks to keep photosynthesis efficient under varying conditions.

• Design challenge:* If you could improve one step of photosynthesis to make it faster, which step would you choose and what change would you make?*

• Further reading* - *Watch the "Plant Physiology - Photosynthetic Adaptations" video series (link) or read Chapter 7 of* Biology Today* (2023 edition).